



Exercise Session 3

📅 06.10.2025

Algorithms & Data Structures HS 25 G-21

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Plan for Today

1. Moodle Quiz

- Moodle quiz

2. Organization

3. Homework

- Discussing exercises from Exercise Sheet 2
- Main focus Ex. 2.1 b, 2.3 a) 4, 2.5 (2.4 if we have time)
- Common mistakes from Exercise Sheet 1

4. Moodle Quiz

- Moodle quiz discussion

5. Theory recap

- Maximum Subarray sum

6. Kahoot!

7. In Class Exercises

- O-notation exercises
- Mathematical Induction

$$n^8 - 800n^7 - 10000n^6 + 200n^2 \leq O(0.25n^7)$$

☐ Wahr

☒ Falsch ✓

$$n^8 - 800n^7 - 10000n^6 + 200n^2 \leq O(0.25n^7)$$

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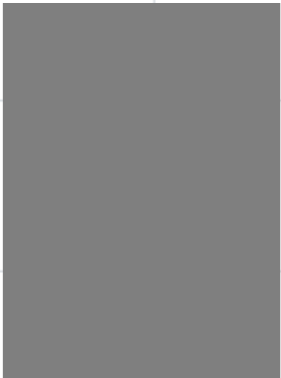
Solution: We will use Theorem 1 to disprove the statement

$$\lim_{n \rightarrow \infty} \frac{n^8 - 800n^7 - 10000n^6 + 200n^2}{0,25 n^7} = \lim_{n \rightarrow \infty} \frac{n^8}{0,25 n^7} - \frac{800n^7}{0,25 n^7} - \frac{10000n^6}{0,25 n^7} + \frac{200n^2}{0,25 n^7} =$$

$$\lim_{n \rightarrow \infty} \frac{n}{0,25} - 800 - \frac{10000}{0,25 n} + \frac{200}{0,25 n^5} = \lim_{n \rightarrow \infty} \frac{n}{0,25} = \infty$$

$$\Rightarrow n^8 - 800n^7 - 10000n^6 + 200n^2 \notin O(0,25 n^7)$$

Let $f(n) = 4e^n$, for which of the following definitions of $g(n)$ is $f(n) \leq O(g(n))$?

True	False			
		$g(n) = n^4 + n^2$		
		$g(n) = n^{100} + e^{100 \ln(n)}$		
		$g(n) = 2^{3n}$		

Bewertung: **Teilpunkte** 

Let $f(n) = 4e^n$, for which of the following definitions of $g(n)$ is $f(n) \leq O(g(n))$?

True	False			
☐ ☒	☒ ☐	$g(n) = n^4 + n^2$	☒	$g(n) = n^4 + n^2$ is polynomial,
☐ ☒	☒ ☐	$g(n) = n^{100} + e^{100 \ln(n)}$	☒	$e^{100 \ln(n)} = n^{100}$ so $g(n) = 2n^{100}$. $f(n) \not\leq O(g(n))$.
☒ ☐	☐ ☒	$g(n) = 2^{3n}$	☒	$2^{3n} = 8^n = e^{\ln(8)n}$, so we can s meaning $f(n) \leq O(g(n))$

Bewertung: **Teilpunkte** ?

Solution: We will use theorem 1.

$$1) \lim_{n \rightarrow \infty} \frac{4e^n}{n^4 + n^2} = \lim_{n \rightarrow \infty} \frac{e^{\ln 4} \cdot e^n}{e^{\ln n^4 + n^2}} = \lim_{n \rightarrow \infty} \frac{e^{\ln 4 + n}}{e^{\ln n^4 + n^2}} = \lim_{n \rightarrow \infty} e^{\ln 4 + n - \ln n^4 - n^2} =$$

$$e^{\lim_{n \rightarrow \infty} \ln 4 + n - 6 \ln n} = e^\infty = \infty \Rightarrow \text{the statement is false}$$

! Polynomial functions always grow asymptotically slower than exponential functions.

Polynomial function example:

$$p(x) = 10x^5 + 8x^4 + 3x^2 + 5 \rightarrow \text{the argument is the base}$$

Exponential function example:

$$q(x) = 21^x \rightarrow \text{the argument is the power}$$

$$2) g(n) = n^{100} + e^{100 \ln(n)} = n^{100} + n^{100} = 2 \cdot n^{100} \Rightarrow g(n) \text{ is a polynomial function, while } f(n) \text{ is an exponential function}$$

$$\Rightarrow f(n) \neq O(g(n))$$

$$3) \lim_{n \rightarrow \infty} \frac{4e^n}{2^{3n}} = \lim_{n \rightarrow \infty} \frac{e^{\ln 4} \cdot e^n}{e^{3n \ln 2}} = \lim_{n \rightarrow \infty} e^{n + \ln 4 - n 3 \ln 2} =$$

$$e^{\lim_{n \rightarrow \infty} n + \ln 4 - n \ln 8} = e^{\lim_{n \rightarrow \infty} n(1 - \ln 8) + \ln 4} = e^{-\infty} = 0$$

$$\Rightarrow f(n) \leq O(g(n))$$

You want to show using induction that a statement $A(n)$ holds for all n such that $n \geq 2$.

Which of the following combinations of *base case* and *induction step* would form a valid proof?

Wählen Sie eine oder mehrere Antworten:

☒ a. *Base case:* $A(2)$ holds.

☒ This is just standard induction.

Induction step: $A(k) \implies A(k+1)$ for all integers $k \geq 2$

☒ b. *Base case:* $A(2), A(3)$ holds.



From base case $n = 2$ we prove it for all positive even numbers and from base case $n = 3$ we prove it for all numbers at least 3, so this covers all the numbers.

Induction step: $A(k) \implies A(k+2)$ for all integers $k \geq 2$.

☐ c. *Base case:* $A(2)$ holds.

Induction step: $A(k) \implies A(k+2)$ for all integers $k \geq 2$.

Solution: a) This is normal induction

b) We have shown $A(2)$ and $A(3)$ hold and the induction step shows that if $A(k)$ holds then $A(k+2)$ holds

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$\Rightarrow$ from $A(2)$, we can show that $A(2r)$ holds $\forall r \geq 1$ and

from $A(3)$, we can show that $A(2r+1)$ holds $\forall r \geq 1$

$\Rightarrow A(n)$ holds $\forall n \geq 2$

c) Because the base case is missing $A(3)$ we can only show that $A(n)$ holds for the even numbers $n \in \mathbb{N} \geq 2$

$$\sum_{i=1}^n i^2 \leq O(n^2 \log(n))?$$

☐ Wahr

☒ Falsch ✓

$$\sum_{i=1}^n i^2 \leq O(n^2 \log(n))?$$

☐ Wahr

☒ Falsch ✓

markieren

Solution: $\sum_{i=1}^n i^2 = \sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} i^2 + \sum_{i=\lceil \frac{n}{2} \rceil}^n i^2 \geq \sum_{i=1}^{\lfloor \frac{n}{2} \rfloor} 1^2 + \sum_{i=\lceil \frac{n}{2} \rceil}^n \left\lceil \frac{n}{2} \right\rceil^2 =$
 $\lfloor \frac{n}{2} \rfloor \cdot 1^2 + (n - \lfloor \frac{n}{2} \rfloor + 1) \left\lceil \frac{n}{2} \right\rceil^2 \geq \left(\frac{n}{2} - 1 \right) \cdot 1^2 + \frac{n}{2} \cdot \left(\frac{n}{2} \right)^2 = \frac{n}{2} - 1 + \frac{n^3}{2} \geq \frac{n^3}{2}$
 $\Rightarrow \sum_{i=1}^n i^2 \geq \frac{n^3}{2}$
 $\Rightarrow \sum_{i=1}^n i^2 \neq O(n^2 \log(n))$
(for $n \geq 2$)

Consider the following recursive formula:

$$T(1) = 0$$

$$T(n) = 2T(n/2) + 5n$$

Assume n is of the form $n = 2^k$ for $k \in \{0, 1, 2, 3, \dots\}$.

Is $T(n) \leq O(n)$?

☐ Wahr

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Assume n is of the form $n = 2^k$ for $k \in \{0, 1, 2, 3, \dots\}$.

Is $T(n) \leq O(n)$?

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Solution:

$$T(1) = 0$$

$$T(n) = 2T(n/2) + 5n = 2 \cdot (2T(n/4) + 5 \cdot \frac{n}{2}) + 5n = 4T(n/4) + 2.5n$$

$$= 4(2T(n/8) + 5 \cdot \frac{n}{4}) + 2.5n = 8T(n/8) + 3.5n = \dots$$

$$= n \cdot 0 + \log n \cdot 5n = 5n \log n$$

$$\Rightarrow T(n) \neq O(n)$$

Thank you for your attention
and participation!